7

Bayesian Cure Model

7.1 Introduction

In additional to the likelihood and frequentist approaches to cure modeling, Bayesian methods have also been applied to survival data with a cure fraction. Ibrahim et al. (2001) provided a comprehensive review of the Bayesian analysis for survival data, and some of the discussions in the review focus on cure rate models, particularly the non-mixture cure model described in Section 2.4.1. The advantage of this type of non-mixture cure model is that the hazard functions are proportional when covariates are used in the analysis. Ibrahim et al. (2001) considered the Bayesian inference for parametric and semiparametric cure models as well as multivariate cure models. The readers are referred to Ibrahim et al. (2001) for detailed discussions of the non-mixture Bayesian cure models. In this chapters, we describe the Bayesian inference including the prior specification, computational methods, posterior distribution, and the model comparison criteria for several alternative parametric and semiparametric cure models. The flexible cure model with latent activation schemes is described in Section 7.2 and the Bayesian cure model with generalized modified Weibull distribution is described in Section 7.3. Section 7.4 presents a Bayesian cure models with spatial random effects. For the three sections, the statistical methods are followed by illustrative examples with BUGS code (Lunn et al., 2009). In Section 7.5, we discuss the general implementation of Bayesian analysis using BUGS language.

7.2 Flexible Cure Model with Latent Activation Schemes

The non-mixture models proposed by Yakovlev et al. (1994b) and Wang et al. (2012a) are motivated by the mechanisms of cancer metastasis. These models assume that an individual is at the risk of failure if a certain number of latent risk factors are activated. Cooner et al. (2007) proposed a unifying class of cure models by accounting for uncertainty in the underlying mechanisms leading to disease manifestation. They called this class of hierarchical models as the flexible cure model under latent activation schemes.

7.2.1 Model Formulation and Inference

Under the latent activation schemes, the observed failure time T is generated by the N latent event times $\tilde{T}_1, \ldots, \tilde{T}_N$. If $N = 0$, then the individual is not exposed to any of the latent events and is considered cured. For a given N, the latent event times $\tilde{T}_1, \ldots, \tilde{T}_N$ are independently identically distributed with a survival function, $P(\tilde{T} > t) = S^H(t) = 1 - F^H(t)$, which is independent of N. A popular choice of parametric distribution is the Weibull distribution, where

$$S^H(t) = \exp\left(-\exp(\mu)t^\sigma\right), \tag{7.1}$$

where μ is the scale parameter and σ is the shape parameter. Let $\tilde{T}_{(1)}, \ldots, \tilde{T}_{(N)}$ be the ordered failure times of $\tilde{T}_1, \ldots, \tilde{T}_N$. Under the general latent activation schemes, r out of N latent factors need to activate for the subject to fail, then $T = \tilde{T}_{(r)}$ for $1 \leq r \leq N$. The rank of the order statistics for the activation of failure r may be a fixed constant, a function of N, or be specified as a random variable conditional on N.

The conditional survival function of $T = \tilde{T}_{(r)}$ given N and r can be written as

$$\begin{aligned} S(t|N,r) &= P(T \geq t|N,r) \\ &= I(N=0) + IB(S^H(t); N-r+1, r) I(1 \leq r \leq N), \end{aligned} \tag{7.2}$$

where $I(\cdot)$ is the indicator function and

$$IB(S^H(t); N-r+1, r) = N \begin{pmatrix} N-1 \\ r-1 \end{pmatrix} \int_0^{S^H(t)} u^{N-r}(1-u)^{r-1} \mathrm{d}u$$

is the incomplete beta function (Cooner et al., 2007). The unconditional survival function $S(t)$ is a mixture of cure ($N = 0$) and the latency distribution for uncured individuals

$$\begin{aligned} S(t) &= E[P(T \geq t|N,r)] \\ &= P(N=0) + E_{N,r}[IB(S^H(t); N-r+1, r) I(1 \leq r \leq N)] \end{aligned} \tag{7.3}$$

where the expectation $E_{N,r}$ is taken over the joint distribution of (N, r). Note that the latency distribution $0 \leq S^H(t) \leq 1$ for any valid distribution of (N, r) with restriction $1 \leq r \leq N$. The cure probability $1 - \pi = P(N = 0)$ only depends on the distribution of N. Usually, the variable N is not observed and parametric distribution may be used. For example, one may assume that $N \sim \text{Poisson}(\lambda)$ with probability density function

$$P(N = k) = \frac{\lambda^k \exp(-\lambda)}{k!}. \tag{7.4}$$

The corresponding cure rate is $P(N = 0) = \exp(-\lambda)$. Other possible distributions for N include Bernoulli, binomial and geometric distributions, and each has its own biological motivations. See Section 2.4 for more discussions.

The marginal distribution of $S(t)$ can be obtained by specifying the joint distribution of r and N through a marginal specification for N and a conditional distribution for r given N. Cooner et al. (2007) considered various latent activation schemes. For example, the first activation scheme is generated by setting $r = 1$. This implies that the event occurs by activating any one of the N latent events, i.e.,

$$T = \tilde{T}_{(1)} = \min\{\tilde{T}_1, \ldots, \tilde{T}_N\}.$$

The last activation scheme is generated by setting $r = N$, where

$$T = \tilde{T}_{(N)} = \max\{\tilde{T}_1, \ldots, \tilde{T}_N\}.$$

Both the first-activation and last-activation schemes may be generated from certain mechanisms of carcinogenics. For certain types of cancer, the first-activation scheme involves metastasis-competent mutation of the tissue mass that generates the first primary cancer. During the short interval of this mutation, the subject's immune response may be activated to initiate N immune responses to this mutation. When there is no mutation, then $N = 0$ and the subject is not susceptible to the cancer. Each immune response is a latent factor capable of resisting disease manifestation or death until its promotion time. Failure occurs after all of the N factors have been activated, so the observed failure time is $T = \max\{\tilde{T}_1, \ldots, \tilde{T}_N\}$, resulting in the last-activation scheme. More generally, a mixture scheme can be generated by a mixture distribution of r given N, where

$$P(r = 1|N) = p \text{ and } P(r = N|N) = 1 - p \tag{7.5}$$

for $0 < p < 1$. The resulting survival function is a mixture of survival functions for the first-activation and last-activation schemes.

In the presence of covariates, one may assume that the parameters μ in equation (7.1) and λ in equation (7.4) are related to covariates. Suppose there are n subjects. For the ith subject, the observed data consist of $(t_i, \delta_i, \boldsymbol{x}_i)$, $i = 1, \ldots, n$, where t_i is the observed failure time, δ_i is the failure indicator, and $\boldsymbol{x}_i$ is a set of covariates. Let μ_i and λ_i be the scale and cure parameters for the ith subject, $i = 1, \ldots, n$. When the covariates are related to the latent event times, one may assume that $\mu_i = \boldsymbol{x}_i^{'}\boldsymbol{\beta}$. When the covariates are related to the cure rate and the complementary log-log link function $\log(\lambda_i) = \boldsymbol{x}_i^{'}\boldsymbol{\gamma}$ is used, the cure rate is given by

$$1 - \pi(\boldsymbol{x}_i) = \log(-\log(\boldsymbol{x}_i^{'}\boldsymbol{\gamma})). \tag{7.6}$$

The logit link function can also be considered for the cure rate model. In the situation that covariates are related to both latent survival times and cure, the regression models for the marginal survival function $S(t)$ may not identifiable. For the issue of identifiability, see further discussions in Cooner et al. (2007).

Let $\boldsymbol{\theta} = (\boldsymbol{\beta}, \boldsymbol{\gamma}, \sigma, \psi)$ denote collection of all parameters to be estimated, where $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ are the regression coefficients for μ_i and λ_i and ψ denotes set

of other hyperparameters that may arise in specific models. The contribution of the ith subject to the likelihood is

$$L(\boldsymbol{\theta}, N_i, r_i) = S(t_i|N_i, r_i)^{1-\delta_i} f(t_i|N_i, r_i)^{\delta_i},$$

where $f(t_i|N_i, r_i) = -\frac{dS(t_i|N_i,r_i)}{dt_i}$ is the corresponding density function of $S(t_i|N_i, r_i)$ in Equation (7.2). The posterior distribution of $\boldsymbol{\theta}$ by marginalizing over N_i and r_i can be written as

$$P(\boldsymbol{\theta}) \propto \prod_{i=1}^{n} \sum_{N_i, r_i} P(\boldsymbol{\theta}, N_i, r_i) \times L(\boldsymbol{\theta}, N_i, r_i), \tag{7.7}$$

where $P(\boldsymbol{\theta}, N_i, r_i)$ are the joint prior probabilities. In general, marginalization in (7.7) is analytically intractable and may be obtained using a MCMC algorithm (Gilks et al., 1995; Carlin and Louis, 2010).

7.2.2 Bayesian Cure Model with Negative Binomial Distribution

As the validity of Poisson distribution for N is questionable (Tucker and Taylor, 1996), de Castro et al. (2009) and Cancho et al. (2011) proposed to use the negative binomial (NB) distribution for N with the probability density function

$$P(N = k|\alpha, \lambda) = \frac{\Gamma(\alpha^{-1} + k)}{\Gamma(\alpha^{-1})k!} \left(\frac{\alpha\lambda}{1 + \alpha\lambda}\right)^k (1 + \alpha\lambda)^{-\frac{1}{\alpha}},$$

where $\lambda > 0, \alpha \geq -1$ and $\alpha > -1/\lambda$.

Under the first-activation scheme, the probability generating function for the NB distribution is $G(s) = \{1 + \alpha\lambda(1 - s)\}^{-1/\alpha}, 0 \leq s \leq 1$. The corresponding marginal survival and density functions are given by

$$\begin{aligned} S(t) &= G(S^H(t)) = \{1 + \alpha\lambda F^H(t)\}^{-1/\alpha}, \\ f(t) &= \{1 + \alpha\lambda F^H(t)\}^{-1/\alpha} \lambda f^H(t). \end{aligned} \tag{7.8}$$

The cure fraction is $1 - \pi = \lim_{t\to\infty} S(t) = (1 + \alpha\lambda)^{-1/\alpha}$. Both the complementary log-log link function and the logit link function can be used in the regression model for the cure rate. Note that the parameter λ is a function of α and cure rate $1 - \pi$, i.e.,

$$\lambda = \begin{cases} [(1 - \pi)^{-\alpha} - 1]/\alpha & \text{if } \alpha \neq 0 \text{ and } \alpha > -1 \\ -\log(1 - \pi) & \text{if } \alpha = 0 \end{cases}.$$

The parameters related to cure rate in model (7.8) are α and $\boldsymbol{\gamma}$, which have intuitive biological interpretations. The parameter λ is the mean number of competing causes and α accounts for the inter-individual variance of the number of causes. For the count data with over-dispersion, the parameter α control

the dispersion. As $\alpha \to 0$, we obtain the Poisson distribution for N and the PHC model or the promotion time cure model (Chen et al., 1999). Under the first-activation scheme, $r = 1$ for all subjects and the likelihood contribution for the ith subject is

$$L(\boldsymbol{\theta}, N_i) = S(t_i|N_i)^{1-\delta_i} f(t_i|N_i)^{\delta_i},$$

where $S(t|N) = S(t|N, r = 1)$ is defined as equation (7.2). A Weibull distribution or a piecewise exponential distribution (Chen et al., 1999) may be used for the latent survival function $S^H(t)$. The piecewise exponential distribution results a semiparametric alternative to the Weibull distribution. Let the time axis be partitioned into L intervals $(0, \tau_1], (\tau_1, \tau_2], \ldots, (\tau_{L-1}, \tau_L]$. If a constant baseline hazard rate λ_j is assumed at the jth interval $(\tau_{j-1}, \tau_j], j = 1, \ldots, L$, the latent survival function can be written as

$$S^H(t) = \exp\left(-\sum_{l=1}^{j-1} \lambda_l(\tau_l - \tau_{l-1}) - \lambda_j(t - \tau_{j-1})\right) \tag{7.9}$$

for $\tau_{j-1} < t \leq \tau_j$. The overall survival function can be written as

$$S(t) = \begin{cases} \{1 + [(1-\pi)^{-\alpha} - 1]F^H(t))\}^{-1/\alpha} & \text{if } \alpha \neq 0 \\ (1-\pi)^{F^H(t)} & \text{if } \alpha = 0 \end{cases}$$

and the density function is

$$f(t) = \begin{cases} \frac{(1-\pi)^{-\alpha}-1}{\alpha} f^H(t)\{1 + [(1-\pi)^{-\alpha} - 1]F^H(t)\}^{-\frac{1}{\alpha}-1} & \text{if } \alpha \neq 0 \\ -\log(1-\pi)(1-\pi)^{F^H(t)} f^H(t) & \text{if } \alpha = 0 \end{cases}.$$

In the presence of covariates $\boldsymbol{x}_i$, the corresponding survival and density functions can be written as $S(t|\boldsymbol{x}_i)$ and $f(t|\boldsymbol{x}_i)$. For the observed survival data $(t_i, \delta_i, \boldsymbol{x}_i), i = 1, \ldots, n$, the loglikelihood function can be written as

$$L(\boldsymbol{\theta}) \propto \prod_{i=1}^{n} f(t_i|\boldsymbol{x}_i, \boldsymbol{\theta})^{\delta_i} S(t_i|\boldsymbol{x}_i, \boldsymbol{\theta})^{1-\delta_i},$$

where $\boldsymbol{\theta} = (\alpha, \boldsymbol{\lambda}, \boldsymbol{\gamma})$, α is the dispersion parameter of the NB distribution, $\boldsymbol{\gamma}$ is the regression coefficients related to cure rate and $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_L)$ are the constant hazard rates for the piecewise exponential distribution. For Bayesian inference, one may assume that $\alpha, \boldsymbol{\gamma}$ and $\boldsymbol{\lambda}$ are independent a priori, i.e.,

$$p(\boldsymbol{\theta}) = p(\alpha)p(\boldsymbol{\gamma})p(\boldsymbol{\lambda}),$$

where $p(\cdot)$ is the prior density functions for the parameters. A discrete uniform prior for α may be used, i.e., $\alpha \in \Omega_\alpha = \{\alpha_1, \ldots, \alpha_A\}$ with probability $1/A$, $A \geq 1$. Yin and Ibrahim (2005a) emphasized that in most real applications there is weak evidence about the specific value of α. If a possible range of α

TABLE 7.1
Specifications of the Bayesian cure models for melanoma survival data

Model	μ	$1-\pi=\exp(-\lambda_i)$	N
1	$\mu_i=\boldsymbol{\beta}'\boldsymbol{x}_i$	$\lambda_i=\lambda$	Poisson(λ)
2	$\mu_i=\mu$	$\log(\lambda_i)=\boldsymbol{\beta}'\boldsymbol{x}_i$	Poisson(λ_i)
3	$\mu_i=\mu$	$\log(\lambda_i)=\boldsymbol{\beta}'\boldsymbol{x}_i$	NB(α,λ_i)

values are used, the final model can be seen as an average of models among the grid Ω_α. The prior distribution for the regression coefficients $\boldsymbol{\gamma}$ may be multivariate normal, i.e., $\boldsymbol{\gamma}\sim MVN(\boldsymbol{\mu}_{\boldsymbol{\gamma}},\Sigma_{\boldsymbol{\gamma}})$. For the piecewise exponential distribution (7.9), let $\xi_j=\log(\lambda_j), j=1,\ldots,L$. The prior distributions for $\xi_1,\ldots,\xi_L$ can be specified as

$$\xi_j|\xi_{j-1},\sigma_j^2\sim N(\xi_{j-1},\sigma_\xi^2), j=1,\ldots,L,$$

and $\sigma_\xi^2\sim IG(a,b)$ with $\xi_0=0$, where $IG(a,b)$ denotes a inverse-gamma distribution with shape parameter a and scale parameter b. Again, the joint posterior density is analytically intractable because the integration of the joint posterior density is not easy to perform. The Bayesian inference based on the MCMC method is described in de Castro et al. (2009).

7.2.3 Application

The melanoma data from the Eastern Cooperative Oncology Group (ECOG) phase III clinical trial E1684 which is used for modeling semiparametric PH mixture cure model (Kirkwood et al., 1996). The survival endpoint is the time to cancer relapse. The covariates include binary variable of sex (0=male, 1=female), binary treatment variable (0=control, 1=IFN treatment) and centered continuous variable for age. In the data set, there are 171 males and 113 females and 140 are in the control group and 144 receive the IFN treatment. The Kaplan-Meier survival curves of the two treatment groups are shown in Figure 7.1. The survival curves of cancer relapse time reach plateaus after six months of treatment, indicating a possibility of cure among the patients.

We apply the Bayesian cure models with different latent activation schemes, the NB distribution for the number of latent events, and the Weibull distribution for the latent distribution. We consider three models and the model specifications are shown in Table 7.1. The three covariates are used in the scale parameter of the Weibull distribution in Model 1 and in the cure probability in Models 2 and 3. Model 1 and 2 consider a mixture of the first-activation and last-activation schemes with the mixing probability p. As recommended by Cooner et al. (2007), we consider a mixture model (7.5) for last-activation and first-activation schemes and use the prior $p\sim U(0,1)$. For Model 1, all patients have the same number of latent events from a Poisson distribution $N\sim$ Poisson(λ), thus the same cure rate $1-\pi=\exp(-\lambda)$ for

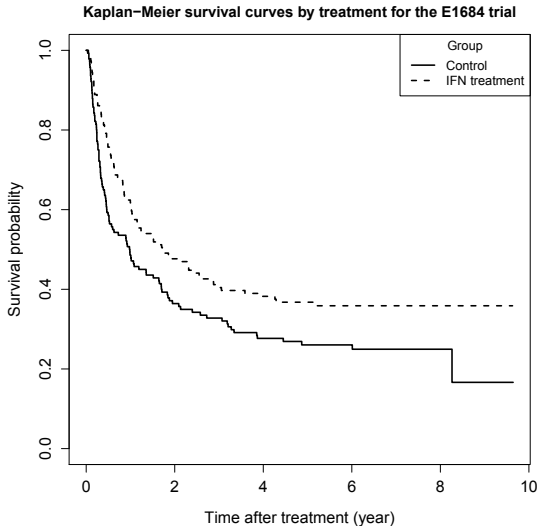

FIGURE 7.1
Kaplan-Meier curves of cancer relapse time for melanoma patients in the E1684 trial.

all subjects. For Model 2, the number of latent events for the ith patient is $N_i = \text{Poisson}(\lambda_i)$ and the cure rates are dependent on the covariates of the subjects. Model 3 considers a last-activation scheme but introduces an extra parameter α in the NB distribution for the number of latent events. For each analysis, we run two initially dispersed parallel MCMC chains for 10,000 iterations each as burn-in, then 25,000 addition runs with thinning factor of 2 to yield $2\times 25{,}000 = 50{,}000$ samples for posterior analysis. The sample OpenBUGS code for fitting Model 3 is as follows:

```
model {
for (i in 1:n) {
  # CDF and pdf
  F[i] <- 1 - exp(-exp(eta)*pow(t[i],gamma))
  f[i] <- exp(eta) * gamma * pow(t[i],gamma-1)) * (1-F[i])

  # Cure rate
  p.cure[i]<-exp(-exp(inprod(beta[1:L],x[i,1:L])))

  # Log-likelihood
  loglik[i]<-(delta[i]*log((pow(p.cure[i],-alpha)-1)/alpha*f[i])
      -(delta[i]+1/alpha)*log(1+(pow(p.cure[i],-alpha)-1)*F[i]))
  zeros[i] ~ dloglik(loglik[i])
}

# Prior distributions
```

```
for (j in 1:K) {
  pi.alpha[j] <- 1/K
}
ind ~ dcat(pi.alpha[1:K])
alpha <- A[ind]
for (j in 1:L) {
  beta[j] ~ dnorm(0,0.01)
}
gamma~dgamma(0.1,0.1)
eta~dnorm(0,0.01)
}
```

The parameter estimates for the three models are shown in Table 7.2. The estimates of mixture proportion p from Models 1 and 2 are above 0.83, indicating that the data is more likely to arise from the first-activation scheme. The estimates of σ from Models 1 and 2 are close to 1, indicating that the exponential distribution may be used for the latent distribution. For Model 1, all patients are assumed to have the same cure rate 0.284. The IFN treatment group has a lower latent hazard rate than the control group with a hazard ratio $\exp(-0.184) = 0.831$ and 95% confidence interval (0.582, 1.192), which is not statistically significant. For Model 2, the covariates are assumed to be related to the cure rate as $1 - \pi(\boldsymbol{x}) = \exp(-\exp(\boldsymbol{x}^{'}\boldsymbol{\beta}))$, and thus a negative coefficient means a higher cure rate for a larger value of a covariate in $\boldsymbol{x}$. For example, the parameter estimate for IFN treatment is -0.393 with 95% confidence interval (-0.697, -0.107), which implies that the patients in the IFN treatment group have statistically significantly higher cure rate than the patients in the control group. The three covariates are assumed to be related to cure fraction in Model 3 in the same way as in Model 2. The parameter estimate of IFN treatment is -0.155 with 95% confidence interval (-0.295, -0.041). Age and sex are not statistically significantly associated with the cure rate in Model 2 and 3.

To evaluate the goodness-of-fit of the three models, we show the estimated survival curves by treatment and sex. We assume that the age for prediction is the average age in the group defined by treatment and sex. For comparison, we also show the Kaplan-Meier curves for cancer relapse by treatment and sex. The plots of the survival curves are shown in Figure 7.2. The dashed, dotted and dash-dotted lines are the predictions from Models 1, 2 and 3, respectively. Model 1 assumes that the cure rates are the same for all patients and Models 2 and 3 assume that the cure rates are associated with treatment and sex. Overall, we see that Models 2 and 3 provide a better fit to the long-term survival rate than Model 1. The estimated survival from Model 3 is between the estimates from Models 1 and 2. Model 3 appears to have a better fit to the early survival than Models 1 and 2. As Cooner et al. (2007) recommended, scientific consideration should guide the final model selection when several models fit the data adequately well. In the E1684 study, treatment, age and

TABLE 7.2
Parameter estimates from the Bayesian cure models with a mixture latent activation scheme

	Model 1				Model 2				Model 3			
Parameter	Mean	SD	95% CI		Mean	SD	95% CI		Mean	SD	95% CI	
Intercept	-0.349	0.155	-0.656	-0.054	0.427	0.125	0.180	0.670	0.319	0.113	0.124	0.581
Treatment	-0.184	0.186	-0.542	0.176	-0.393	0.151	-0.697	-0.107	-0.155	0.064	-0.295	-0.041
Sex	0.146	0.183	-0.206	0.505	-0.048	0.153	-0.353	0.243	0.008	0.057	-0.110	0.116
Age	-0.007	0.006	-0.020	0.005	0.006	0.006	-0.005	0.018	0.001	0.002	-0.003	0.006
Cure $1-\pi$	0.284	0.028	0.234	0.342								
μ					-0.357	0.126	-0.597	-0.100	-4.350	1.490	-8.524	-2.425
p (α)	0.861	0.108	0.599	0.995	0.838	0.108	0.584	0.990	6.398	1.411	4.000	9.000
σ	1.009	0.060	0.892	1.127	1.014	0.057	0.901	1.124	2.244	0.336	1.651	2.964

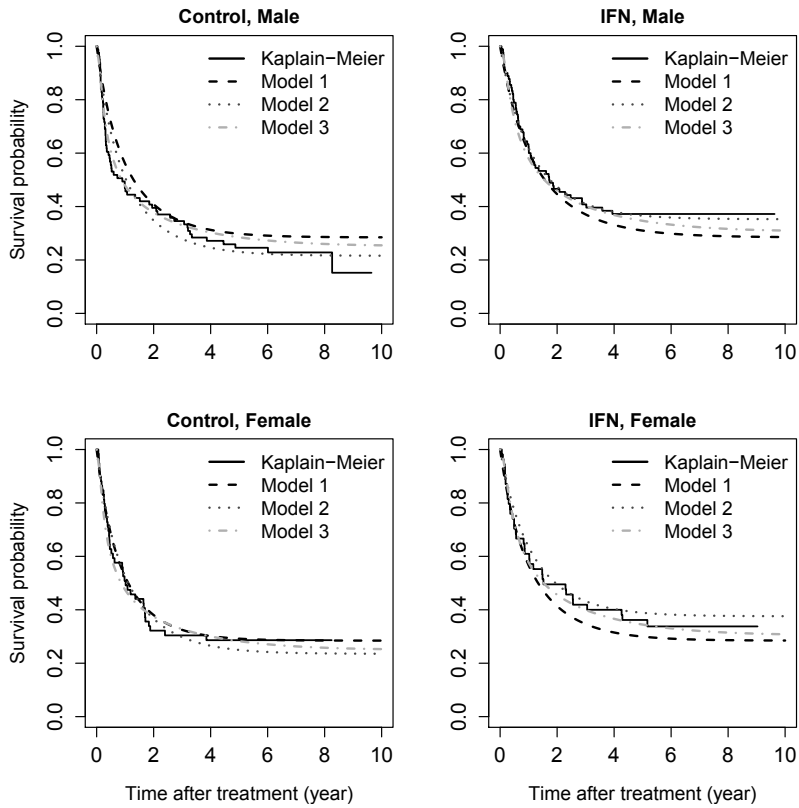

FIGURE 7.2
Comparison of the predicted survival from the three models for the E1684 study.

sex are significant prognostic factor of cure, therefore Models 2 and 3 that assume a covariate-dependent cure rate are more scientifically appropriate.

7.3 Bayesian Cure Models with Generalized Modified Weibull Distribution

7.3.1 Model Formulation and Inference

We introduce here a Bayesian analysis for the cure models based on the generalized modified Weibull (GMW) distribution, which is a very general class of

TABLE 7.3
Parametric distributions as special cases of the GMW distribution

Distribution	Parameter restrictions
Weibull	$\delta = 0, \beta = 1$
Exponential	$\delta = 0, \beta = 1, \eta = 1$
Rayleigh	$\delta = 0, \beta = 1, \eta = 2$
Type I Extreme value	$\beta = 1, \eta = 0$
Exponentiated Weibull	$\delta = 0$
Exponentiated Exponential	$\delta = 0, \eta = 1$
Generalized Rayleigh	$\delta = 0, \eta = 2$
Modified Weibull	$\beta = 1$

distributions and easy to implement (Martinez et al., 2013). The mixture cure model is given by (2.1) with the GMW distribution as the latency distribution with the following survival and density functions:

$$S_u(t) = 1 - \{1 - \exp[-\alpha t^\eta \exp(\delta t)]\}^\beta,$$
$$f_u(t) = \frac{\alpha\beta t^{\eta-1}(\eta + \delta t)\exp[\delta t - \alpha t^\eta \exp(\delta t)]}{1 - \exp[-\alpha t^\eta \exp(\delta t)]^{1-\beta}},$$

where $\alpha > 0, \beta > 0, \eta \geq 0$ and $\delta \geq 0$, but η and δ cannot be 0 simultaneously. The GMW distribution is denoted by $GMW(\alpha, \beta, \eta, \delta)$. The parameter α is a scale parameter and β and η are shape parameters. The four-parameter distribution was introduced by Carrasco et al. (2008) and it is flexible to accommodate many forms of the hazard rate function, including bathtub-shaped hazard rate function. Table 7.3 shows several standard parametric distributions as the special cases of the four-parameter GMW distribution. The corresponding density function of the GMW distribution is given by

Assume that $\pi_c(\boldsymbol{x}) = 1 - \pi(\boldsymbol{x})$ and

$$\pi_c(\boldsymbol{x}) = \frac{\exp(\boldsymbol{x}'\boldsymbol{\gamma})}{1 + \exp(\boldsymbol{x}'\boldsymbol{\gamma})}, \tag{7.10}$$

The overall survival function of the mixture cure model is

$$S(t|\boldsymbol{x}) = \pi_c(\boldsymbol{x}) + (1 - \pi_c(\boldsymbol{x}))S_u(t),$$

For observed survival data $(t_i, \delta_i, \boldsymbol{x}_i), i = 1, \ldots, n$, the likelihood function for the mixture cure model is given by

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{n}[(1 - \pi_c(\boldsymbol{x}_i))f_u(t_i)]^{\delta_i}[\pi_c(\boldsymbol{x}) + (1 - \pi_c(\boldsymbol{x}))S_u(t)]^{1-\delta_i}. \tag{7.11}$$

For the non-mixture cure model, the survival function is given by

$$S(t|\boldsymbol{x}) = \{\pi_c(\boldsymbol{x})\}^{F_u(t)} = \exp\{\log(\pi_c(\boldsymbol{x})F_u(t))\},$$

where $F_u(t) = 1 - S_u(t)$ is a proper cumulative distribution function. Unlike the mixture cure model, $F_u(t)$ cannot be interpreted as the cumulative distribution function for the uncured patients. The likelihood function for the non-mixture cure model is given as

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{n} \{-\log(\pi_c(\boldsymbol{x})F_u(t))\}^{\delta_i} \exp\{\log(\pi_c(\boldsymbol{x})F_u(t))\}. \tag{7.12}$$

For a Bayesian analysis of the both types of cure models, we assume gamma and half-normal (HN) prior distributions for the parameters α, β, η and δ considering the fact that these parameters are real and positive numbers:

$$\alpha \sim \text{Gamma}(a_\alpha, b_\alpha), \quad \beta \sim \text{Gamma}(a_\beta, b_\beta),$$
$$\eta \sim \text{HN}(a_\eta, b_\eta), \qquad \delta \sim \text{HN}(a_\delta, b_\delta),$$

where a_α, b_α, a_β, b_β, a_η, b_η, a_δ and b_δ are known hyperparameters and $\text{Gamma}(a, b)$ denotes a Gamma distribution with mean a/b and and variance a/b^2.

It is important to note that one or more of these parameters may be fixed with constant values when considering the special cases of the GMW distribution. For example, if considering the exponentiated exponential distribution, $\eta = 1$ and $\delta = 0$. For the regression coefficients $\boldsymbol{\gamma}$ for cure rate, we assume a multivariate normal distribution where $\boldsymbol{\gamma} \sim MNV(0, \Sigma_\gamma)$.

7.3.2 Application

As an illustration, we apply the Bayesian mixture and non-mixture cure models with the GMW distribution to survival data from an International Bone Marrow Transplant Registry study of alternative donor bone marrow transplants (BMT) reported in Szydlo et al. (1997). As in many BMT studies, there are two competing risks of treatment failure in this study; relapse (n=311/1715 cases), i.e., recurrence of the primary disease, and death in complete remission (n=557/1715), also known as treatment-related mortality. This study consisted of patients with acute lymphoblastic leukemia (ALL) (n=537), acute myelogenous leukemia (AML) (n=340), or chronic myelogenous leukemia (CML) (n=838). Patients were transplanted with their disease in an early (n=1026), intermediate (n=410), or advanced (n=279) stage based on remission status. The initial Karnofsky performance score was greater than or equal to 90 in 1382 cases. Of primary interest is the comparison of relapse and treatment-related mortality between patients with different donor types. There were 1224 with a human leukocyte antigen (HLA)-identical sibling donor, 383 with an HLA-matched unrelated donor, and 108 cases with an HLA-mismatched unrelated donor. The data were analyzed as an example of competing-risk analysis in Klein and Andersen (2005).

We restricted the analysis to the patients with CML and considered the event as relapse or treatment-related mortality. Therefore, the cured patients

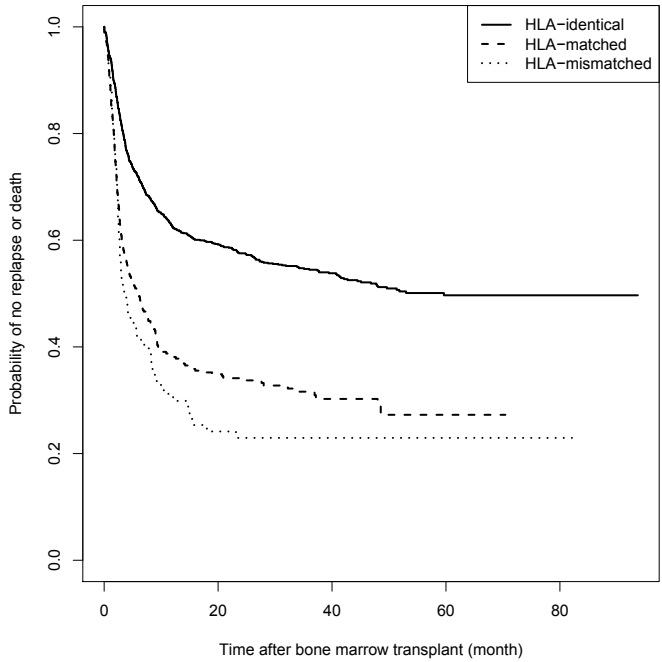

FIGURE 7.3
Kaplan-Meier survival curves for the CML patients with 3 donor types.

are those who did not have either event. The Kaplan-Meier survival curves for the 3 donor groups are shown in Figure 7.3. The survival curves clearly reach a plateau in 60 months after bone marrow transplant. The patients with HLA-identically matched donors have the highest long-term survival rates.

We fit the Bayesian cure models with the GMW distribution and its special cases listed in Table 7.3 to the BMT data. Figure 7.3 shows that the HLA donor type is a strong predictor of long-term survival rates. Thus, we use it as a categorical covariate in the incidence model (7.10) for the cure rate. We also consider covariates in the scale parameter α as $\alpha(\boldsymbol{x}) = \exp(\beta_\alpha \boldsymbol{x})$. Following Carrasco et al. (2008), we use the Deviance Information Criterion (DIC) for model selection. Models with smaller DIC values are better supported by the data. The DIC values for the various cure models are shown in Table 7.4. The model with the smallest DIC 3340 is the mixture cure model with exponentiated Weibull distribution and the covariate donor type is only used for the cure rate in this model. The WinBUGS code for fitting this model is as follows:

```
model {
for (i in 1:N) {
```

TABLE 7.4
DIC values of the Bayesian mixture and non-mixture cure models with different latency distributions

Use of covariate	π_c only		π_c and α	
Latency distribution	Mixture	Non-mixture	Mixture	Non-mixture
GMW	3364	3343	3370	3344
Weibull	3413	3384	3390	3380
Exponential	3435	3386	3402	3379
Rayleigh	4288	4081	4161	4001
Type I Extreme Value	4322	4276	4288	4268
Exponentiated Weibull	3361	3340	3349	3341
Exponentiated Exponential	3428	3387	3400	3381
Generalized Rayleigh	3510	3449	3475	3437
Modified Weibull	3414	3386	3392	3381

```
  f0A1[i]<-alpha*beta*pow(t[i],eta-1)*(eta+delta*t[i])
  f0A2[i]<-exp(delta*t[i]-alpha*pow(t[i],eta)*exp(delta*t[i]))
  f0B[i]<-pow(1-exp(-alpha*pow(t[i],eta)*exp(delta*t[i])),1-beta)
  f0[i]<-(f0A1[i]*f0A2[i])/f0B[i]
  S0[i]<-1-pow(1-exp(-alpha*pow(t[i],eta)*exp(delta*t[i])),beta)
  logit(p[i])<-inprod(gamma[1:K],x[i,1:K])
  F0[i] <- 1-S0[i]
  h[i] <- -(log(p[i]))*f0[i]
  L[i] <- pow(h[i],d[i])*exp(F0[i]*log(p[i]))
  logL[i] <- log(L[i])
  zeros[i] ~ dloglik(logL[i])
}

for (i in 1:K) {
  gamma[i]~dnorm(0,0.01)
}

alpha ~ dgamma(1,1)
beta ~ dgamma(1,1)
eta ~ dnorm(0,0.01)I(0,)
delta<-0
}
```

We present the model fit for this model in Figure 7.4. We see the model fits the data well and the cure rates are 57.2%, 24.8% and 17.2% for the patients with identical, matched and mismatched donors, respectively.

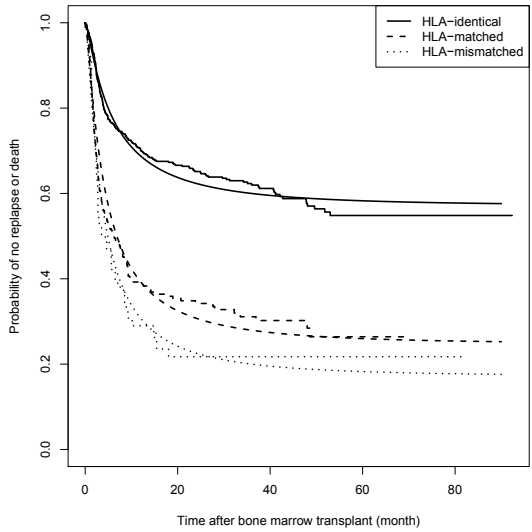

FIGURE 7.4
Estimated survival curves from the Bayesian cure model for the BMT data.

7.4 Bayesian Mixture Cure Model with Spatially Correlated Frailties

7.4.1 Spatial Mixture Cure Model

Bayesian methods have been used to fit the cure model with random effects (frailty). For example, de Souza et al. (2017) used a discrete probability distribution including zero for the frailty in the mixture cure model. Rahimzadeh et al. (2010) used two different types of frailty with the bivariate log-normal distribution instead of the gamma distribution in the non-mixture cure models. Cooner et al. (2006) proposed a non-mixture cure model with spatial frailties for geographically referenced survival data and Banerjee and Carlin (2004) developed a parametric spatial cure rate models for interval-censored survival data from smoking cessation trials. Seppa et al. (2010) and Yu and Tiwari (2012) proposed similar mixture cure models with spatial frailties for population-based cancer survival data. In this section, we briefly describe the mixture cure model with spatially correlated frailties for population-based cancer survival data.

The survival function of the mixture cure model is specified as

$$S(t|\boldsymbol{x}) = \pi_c(\boldsymbol{x}) + (1 - \pi_c(\boldsymbol{x}))S_u(t|\boldsymbol{x}), \tag{7.13}$$

where the cure fraction $\pi_c(\boldsymbol{x})$ is modeled as in (7.10). For the sake of simplicity,

we use the same set of covariates $\boldsymbol{x}$ in both the latency and incidence parts of the mixture cure model. The latency distribution $S_u(t|\boldsymbol{x})$ may be parametric or semiparametric. Seppa et al. (2010) considered a generalized gamma distribution and Yu and Tiwari (2012) considered a mixture of Weibull, lognormal and loglostic distributions for the latency part. For illustration, we consider the Weibull distribution with the survival function given by

$$S_u(t|\boldsymbol{x}) = \exp\left\{-\exp\left[\mu(\boldsymbol{x})\right] t^{\sigma}\right\},$$

where μ is the rate parameter and σ is the scale parameter. The median survival time is $[\exp(-\mu)\log(2)]^{\frac{1}{\sigma}}$. We further assume that the rate or location parameter is related to covariates as

$$\mu(\boldsymbol{x}) = \boldsymbol{x}^{'}\boldsymbol{\beta}. \tag{7.14}$$

For population-based cancer survival data, we will focus on the cause-specific survival due to cancer, where the death due to cancer is treated as event while loss to follow-up and death due to other causes are treated as censoring. Because of the large sample size, the population-based cancer survival data may be stratified by variables like age, calendar year and cancer stages at diagnosis. In addition, the survival times after diagnosis are grouped into monthly or annual intervals $I_j = [t_{j-1}, t_j), j = 1, 2, \ldots, J$, where t_J is the study cutoff time. For the stratum with covariates $\boldsymbol{x}$, let n_{xj} be the number of people alive at the beginning of interval I_j. For cause-specific cancer survival analysis, we consider d_{xj} as the number of patients dying from cancer of interest and l_{xj} as the number of patients lost to follow-up or dying from other causes in interval I_j. Using the actuarial assumption, the adjusted number of person-years at risk is $r_{xj} = n_{xj} - 0.5l_{xj}$. The probability of dying from cancer during the interval I_j given that a subject is alive at the beginning of the interval is given by

$$p_j(\boldsymbol{x}) = P(T < t_j | T \geq t_{j-1}; \boldsymbol{x}) = 1 - \frac{S(t_j|\boldsymbol{x})}{S(t_{j-1}|\boldsymbol{x})}, \tag{7.15}$$

where $S(t|\boldsymbol{x})$ is the survival function defined in (7.13).

The number of deaths due to cancer in an interval is usually assumed to follow a binomial distribution, i.e., $d_{xj} \sim \text{Binomial}(r_{xj}, p_j(\boldsymbol{x}))$. When the cancer has a good prognosis or curable, the number of deaths due to cancer may be sparse or even zero in late intervals, especially for small geographical regions in spatial analysis. In this situation, the binomial distribution may not be appropriate and a Poisson distribution may be used, i.e.,

$$d_{xj} \sim \text{Poisson}(r_{xj} p_j(\boldsymbol{x})).$$

Let $\boldsymbol{\theta} = (\boldsymbol{\beta}, \boldsymbol{\gamma}, \sigma)$ be the vector of all parameters to be estimated. The corresponding likelihood function is written as

$$L(\boldsymbol{\theta}) = \prod_{\boldsymbol{x}} \prod_{j=1}^{J} (r_{xj} p_j(\boldsymbol{x}))^{dxj} \exp(-r_{xj} \times p_j(\boldsymbol{x})). \tag{7.16}$$

The results from the two distributional assumptions for the number of deaths are usually very similar since the log-likelihoods are similar for survival data with moderate or low annual death rates (Dickman et al., 2004).

The spatial pattern and the temporal trends of cancer survival are gaining more interest recently, and health policy makers and public health researchers started to investigate the spatial association and patterns of health outcomes. Suppose that there are I geographical regions and let $\boldsymbol{r}_i = (r_{i1}, r_{i2})$ be the bivariate random effects (frailties), where r_{i1} denotes the frailty for the cure fraction and r_{i1} denotes the frailty for the survival for the uncured patients. Similar to the models discussed in Section 4.3.1, the cure fraction and the location parameter for the uncured patients for the ith geographic region can be written as

$$\pi_{ci}(\boldsymbol{x}) = \frac{\exp(\boldsymbol{x}'\boldsymbol{\gamma} + r_{i1})}{1 + \exp(\boldsymbol{x}'\boldsymbol{\gamma} + r_{i1})} \tag{7.17}$$

and

$$\mu_i(\boldsymbol{x}) = \boldsymbol{x}'\boldsymbol{\beta} + r_{i2}, \tag{7.18}$$

respectively. The multivariate conditionally autoregressive (CAR) distributions can be used for the spatial random effects in the baseline hazard rate and the spatial frailties. Most CAR models are members of the family developed by Mardia (1988). Specifically, the multivariate CAR of Gamerman et al. (2003) generalizes the univariate pairwise difference joint prior as

$$\pi(\boldsymbol{r}_1, \ldots, \boldsymbol{r}_I | \Omega_r) \propto |\Omega_r|^{-I/2} \exp\left\{ \sum_{i \neq j}^{I} c_{ij} (\boldsymbol{r}_i - \boldsymbol{r}_j)' \Omega_r^{-1} (\boldsymbol{r}_i - \boldsymbol{r}_j) \right\},$$

where $\boldsymbol{r}_i = (r_{i1}, \ldots, r_{ip})', i = 1, \ldots, I$, be the multivariate p-dimensional ($p \geq 2$, $p = 2$ in the mixture cure model above) vector of spatially correlated Gaussian random effects in the I areas, Ω_r is the dispersion matrix and $c_{ij} = 1$ if area i and j are adjacent and 0 otherwise (Congdon, 2007). The intrinsic multivariate CAR prior with 0-1 adjacency weights (Besag et al., 1991) gives the conditional distribution

$$\boldsymbol{r}_i | \boldsymbol{r}_{(-i)} \sim \mathrm{MVN}_p(\bar{\boldsymbol{r}}_i, V/n_i),$$

where $\boldsymbol{r}_{(-i)}$ denotes the elements of the $2 \times I$ matrix excluding the ith area (column), $\bar{\boldsymbol{r}}_i = (\bar{r}_{i1}, \bar{r}_{i2})$ with $\bar{r}_{ip} = \sum_{j \in \delta_i} r_{jp}/n_i$, δ_i and n_i denote the set of labels of the neighbors of area i and the number of neighbors, respectively, and V is a 2×2 dispersion matrix. The intrinsic multivariate CAR is currently implemented in GeoBUGS, a module of WinBUGS (Lunn et al., 2009). The syntax for a CAR distribution is

```
R[1:p,1:J] ~ mv.car(adj[], weights[], num[], omega[,])
```

where `adj[]` is a vector that represents the adjacency matrix for the study region, `weights[]` gives unnormalized weights associated with each pair of

areas, `num[]` gives the number of neighbors for each area and `omega[,]` is the precision matrix.

Let $p(\boldsymbol{\theta})$ be the joint prior distribution of $\boldsymbol{\theta}$ and let $L(\boldsymbol{\theta})$ be the likelihood function for the grouped cause-specific survival data given by (7.16). The joint posterior distribution of $\boldsymbol{\theta}$ is

$$p(\boldsymbol{\theta}) \propto L(\boldsymbol{\theta})p(\boldsymbol{\theta}).$$

We assume that the prior distributions for the parameters are mutually independent, i.e.,

$$p(\boldsymbol{\theta}) = p(\boldsymbol{\beta})p(\boldsymbol{\gamma})p(\sigma)p(\boldsymbol{r}),$$

where $p(\boldsymbol{\beta})$ and $p(\boldsymbol{\gamma})$ are the priors for regression coefficients, $p(\sigma)$ is the prior for the scale parameter and $p(\boldsymbol{r}) = p(\boldsymbol{r}_1, \ldots, \boldsymbol{r}_I|\Omega_r)$ is the prior for the spatial random effects. In particular, we assume conjugate prior for the model parameters: $p(\boldsymbol{\beta}) \sim MVN(\mu_{\boldsymbol{\beta}}, \Sigma_{\boldsymbol{\beta}}), p(\boldsymbol{\gamma}) \sim MVN(\mu_{\boldsymbol{\gamma}}, \Sigma_{\boldsymbol{\gamma}})$, $p(\sigma^2) \sim IG(a, b)$, where $IG(a, b)$ denotes an inverse gamma distribution with shape parameter a and scale parameter b. In addition, we also assume that the hyperprior for Ω_r is an inverse Wishart distribution with scale matrix C_γ and ν_γ degrees of freedom.

We use the MCMC algorithm for parameter estimation and inferences. The proposed method can be implemented in the freely available software WinBUGS or OpenBUGS (Lunn et al., 2009). After a sufficient number of burn-in iterations, we use the remaining samples from the MCMC simulations to estimate any function of the parameters of interest. In order to see how stable the final estimates are, multiple MCMC runs are conducted with different initial values and starting points. The convergence of the MCMC samples of the parameters after excluding the initial burn-in samples are monitored using the R package `CODA` (Best et al., 1995). For example, Gelman and Rubin (1992) used a 'potential scale reduction factor' for each parameter in $\boldsymbol{\theta}$, together with upper and lower confidence limits.

7.4.2 Application

We apply the spatial mixture cure model to estimate the survival and cure rate for patients diagnosed with lung and bronchus cancer from 1992 to 2013 in Iowa. The survival data was extracted from the Surveillance, Epidemiology, and End Results (SEER) 13 registries and the cutoff date for survival analysis is the end of 2013. The relative survival data are stratified by SEER historical stage and 99 counties in Iowa. The SEER historical stage is a summary stage that categorize how far a cancer has spread from its point of origin. It is a combination of the most precise clinical and pathological documentation of the extent of disease. The SEER Program classifies the historical stages into Localized, Regional and Distant stages consistently for all cancer sites for the appropriate years. The maximum follow-up time is 21 years, with sufficient time to observing the possible plateau of survival curve if there is a cure. The

survival times are grouped into annual intervals. The survival data include the number of subjects being alive in the beginning of the interval, the number of subjects dying from lung and bronchus cancer and the number of loss of follow-up during each of the intervals. The WinBUGS code for fitting the Bayesian spatial cure model is as follows:

```
model {
for (i in 1:Nx) {
  # probability and mean number of deaths in each intervals
  logit(p.cure[i])<-inprod(alpha[1:3],x[i,1:3])+R[1,G[i]]
  mu[i]<-inprod(beta[1:3],x[i,1:3])+R[2,G[i]]
  for (j in 1:maxT) {
    S[i,j]<-p.cure[i]+(1-p.cure[i])*exp(-exp(mu[i])*pow(j,sigma))
  }
  p[i,1]<-(1-S[i,1])*n[i,1]
  for (j in 2:nFup[i]) {
        p[i,j]<-(1-S[i,j]/S[i,j-1])*n[i,j]
  }

  # poisson model for number of deaths
  for (j in 1:nFup[i]) {
        d[i,j]~dpois(p[i,j])
  }
}

# priors for alpha, beta and sigma
alpha[1]~dnorm(0,0.01)
beta[1]~dnorm(0,0.01)
for (i in 2:3) {
  alpha[i]~dnorm(0,0.01)I(,0)
  beta[i]~dnorm(0,0.01)I(0,)
}
sigma~dexp(1)

# priors for spatial random effect
for (i in 1:nsum) {
  weights[i]<-1
}
R[1:2,1:M] ~ mv.car(adj[], weights[], num[], tau[,])
tau[1:2,1:2] ~ dwish(A.tau[,],2)
}
```

In the analysis, the total number of patients diagnosed with lung and bronchus cancer is 212,619. The historical stage is used as a categorical covariates in both cure fraction and latency distribution as specified in (7.10) and (7.14). The posterior mean, standard deviation (SD) and 95% confidence interval (CI) of the parameters are shown in Table 7.5. The reference group is the patients with localized cancer, the cure rate estimate is 62.7% with 95% CI (58.5%, 69.8%). Compared to the patients with localized cancer, the patients

TABLE 7.5
Parameter estimates for the spatial mixture cure model for lung and bronchus cancer survival data

	Latency distribution			Cure		
Variable	Mean	SD	(95% CI)	Mean	SD	(95% CI)
Intercept	-2.133	0.067	(-2.264,-2.002)	0.520	0.091	(0.343, 0.698)
Localized						
Regional	1.101	0.068	(0.968, 1.235)	-0.679	0.086	(-0.848, -0.510)
Distant	2.060	0.069	(1.924, 2.196)	-2.449	0.091	(-2.626, -2.271)
σ	0.639	0.005	(0.628, 0.649)			

TABLE 7.6
Mean, minimum and maximum cure fraction and median survival time for the uncured patients.

Stage	Median survival for uncured			Cure fraction (%)		
	Minimum	Mean	Maximum	Minimum	Mean	Maximum
Localized	12.7	15.9	19.6	56.2	62.7	68.9
Regional	2.3	2.8	3.5	39.4	46.1	52.9
Distant	0.5	0.6	0.8	10.0	12.7	16.1

with regional and distant cancers have significantly higher hazard of dying from lung and bronchus cancer and significantly lower cure rates. Therefore, the cancer stage is a significant prognostic factor for cancer survival. This also implies the importance of early diagnosis for survival. Table 7.6 shows the mean, minimum and maximum of the cure fractions and median survival times for the uncured patients for the 99 counties by SEER historic stage.

The mean cure rate for the distant lung and bronchus cancer for the 99 counties is 12.7% with a range of 10.0%–16.1%. The cure rates for the localized cancer have a wider range of 56.2%–68.9% and mean 62.7%. We show the map of the county-level cure rates for localized lung and bonchus cancer in Figure 7.5. From this map, we see that the northern Iowa counties tend to have slightly higher cure rates than the southern counties. This is probably due to the differences in patient characteristics or health care quality in this area. It provides useful information to health care planners for allocating limited resources for cancer control.

7.5 Implementation

We consider the example of the Bayesian cure model with latent activation schemes, discussed in Section 7.2. The survival data include cancer relapse

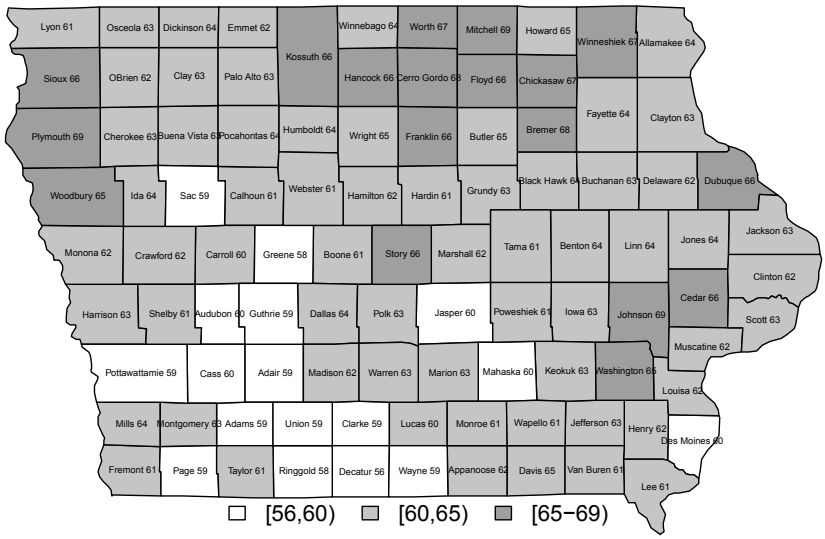

FIGURE 7.5
Map of cure rate (%) for localized lung and bronchus cancer in Iowa.

time `y` and event indicator `d` for n patients, and the data matrix `x` include `K` columns for $K-1$ covariates and one column of constant 1. The parameters to be estimated are $\boldsymbol{\beta}, \theta, \sigma$ and p. The BUGS syntax code for the Bayesian cure model is specified in Section 7.4.2.

Bayesian cure model has been implemented using the OpenBUGS, which can be called within R using BRugs (Thomas et al., 2006). BRugs is a collection of R functions that allow users to analyze graphical models using MCMC techniques. BRugs uses the same model specification language as WinBUGS or OpenBUGS and the same format for data and initial values. The BRugs functions can be split into two groups: those associated with setting up and simulating the graphical model and those associated with making statistical inference. In general, the R functions in BRugs correspond to the command buttons and text entry fields in the menus of WinBUGS. The example syntax of using BRugs is shown below:

```
library("BRugs")
# Setting up the data list and initial parmaters
data.list<-list(n=n,y=y,x=x,d=d,K=K)
parm.init1<-list(beta = rep(0,4), theta = 0.5, sigma = 1.0, p=0.5)
parm.init2<-list(beta = rnorm(4), theta = 0.4, sigma = 1.0, p=0.2)

# Create files for data, initial parameters
# Create data file
bugsData(data.list, fileName="bugsData.txt")
```

```
# Initial parameters 1
bugsData(parm.init1, fileName="bugsInit1.txt")
# Initial parameters 2
bugsData(parm.init2, fileName="bugsInit2.txt")

# Bayesian inference with MCMC
modelCheck("BayesianCureModel.txt")  # Check model file
parm<-c("beta","theta","p","sigma")      # Parameter list
modelData("bugsData.txt")            # Read data file
modelCompile(numChains=2)            # Compile model with 2 chains
modelInits("bugsInit1.txt")          # Read initial for 1st chain
modelInits("bugsInit2.txt")          # Read initial for 2nd chain
modelSetRN(1)                        # Set random seed
modelUpdate(5000)                    # Burn in 5000 runs
samplesSet(parm)                     # Set parameters
# 10,000 more runs with thinning factor 2
modelUpdate(10000,thin=2)

# output parameter estimates and plots
# Output estimates to a data frame
parm.estimate=samplesStats("*")
# plot the sample history
samplesHistory("*",mfrow=c(2,4),ask=FALSE)
# plot the sample density
samplesDensity("*",mfrow=c(2,4),ask=FALSE)
# Auto-correlation of beta in 1st chain
samplesAutoC("beta",chain=1)
```

In the set-up stage, the analysis data file is generated from a data list `data.list` and two files for initial parameters are generated from two data lists. After the MCMC simulations, the parameter estimates can be saved into a data frame `parm.estimate`. An alternative way to call WinBUGS/OpenBUGS is to use the R package `R2WinBUGS`:

```
library("R2WinBUGS")
data.list<-list(n=n, y=y, x=x, d=d,K=K)
inits<-list(parm.init1, parm.init2)
parm<-c("beta","theta","sigma","p")

BayesianCureModel<-bugs(
 data.list,
 inits,
 model.file="BayesianCureModel.txt",
 parameters.to.save=parm,
 n.chains=2,
 n.iter=20000,
 n.burnin=10000,
 n.thin=2,
 bugs.directory="C:/WinBUGS/winbugs14_full_patched/WinBUGS14",
 working.directory="C:/Cure Model Book/chapter7",
```

```
  codaPkg=TRUE,
  debug=TRUE
 )
```

The input data set `data.list` is either a named list with names corresponding to variable names in the model file or a list of the names of the data objects used by the model. The initial value `inits` is a list with `n.chains` elements; each element of the list is itself a list of starting values for the model, or a function creating possibly random initial values. Alternatively, if `inits=NULL`, initial values are generated by WinBUGS. If `inits` is a character vector with `n.chains` elements, it is assumed that `inits` have already been written to the working directory. The vector of parameters `parm` includes the parameters to be monitored. The arguments `n.chain`, `n.iter`, `n.burnin` and `n.thin` are the number of chains, number of MCMC iterations, burn-ins and thinning factors, respectively. Readers are referred to Sturtz et al. (2005) for details.

7.6 Summary

Bayesian analysis of cure models is increasingly used because of the convenience of the MCMC method. In this chapter, we described several extensions and implementation of the mixture and non-mixture cure models. Non-informative or weakly informative priors are used to reduce the impact of priors. We consider using DIC as the goodness-of-fit measure for model selection. For the cure model with latent activation schemes, Cooner et al. (2007) mentioned that the DIC may not be appropriate because that the likelihood function is not log-concave and suggested using the L-measure. Furthermore, it is challenging to determine how to use the covariates in the regression for the parameters in the cure models. In practice, the goodness-of-fit from different models may be very close, and scientific knowledge should be used to guide how to pick the covariates and select the final models.

The Bayesian framework is easy to implement and can be extended to more general situations, e.g., the Bayesian cure model with frailty (de Souza et al., 2017) and Bayesian cure model that accommodates both multiplicative and additive covariates (Nieto-Barajas and Yin, 2009). Bremhorst and Lambert (2016) proposed a flexible estimation method using Bayesian P-splines for cure models.

Recently, the Bayesian methods are increasingly used in clinical trials. For example, informative priors that incorporate historical data or expert opinions are necessary because of limited data. Bayesian design that use historical data may be used for clinical trials for pediatric populations or rare indications with cure rate (Psioda and Ibrahim, 2018). It is of interest to explore how to use power prior to utilize historical data to reduce the number of patients in the control group (Chen et al., 2002).

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